APPENDIX

Cardiac Dynamics: Mathematical Study of Cardiomyocyte Models





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A Summary of Fenichel Theorems

The most important results with application in the context of this work have already been mentioned, but here a short summary of the main theorems will be provided. We will follow the notation and statements presented in Kuehn's book[1].
In the thesis, a general statement of Fenichel Theory has been presented with a sketch of the proof.
Following the usual notation for fast-slow systems, we have a system with a function $f : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^m$ for the fast variables and a function $g : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$ for the slow variables.
Notation: Let X_ε denote the vector field defined on the fast time scale:

$$\begin{cases} x' = f(x,y,\varepsilon), \\ y' = \varepsilon g(x,y,\varepsilon). \end{cases} \tag{1}$$

That is, $X_\varepsilon(x,y) = (f(x,y,\varepsilon), \varepsilon g(x,y,\varepsilon))$.
In particular, when $\varepsilon = 0$, we obtain the system

$$\begin{cases} x' = f(x,y,0), \\ y' = 0. \end{cases} \tag{2}$$

$X_0(x,y) = (f(x,y,0),0)$ is the vector field that defines the fast flow.
Analogously, we consider the system in the slow time scale:

$$\begin{cases} \varepsilon \dot{x} = f(x,y,\varepsilon), \\ \dot{y} = g(x,y,\varepsilon). \end{cases} \tag{3}$$

In the singular limit, this turns into the system

$$\begin{cases} 0 = f(x,y,0), \\ \dot{y} = g(x,y,0). \end{cases} \tag{4}$$

whose flow is called the slow flow and will be denoted by X_R.

A.1 Classical Fenichel Theorem

Now we introduce some classical statements of the Fenichel Theorem.

Theorem A.1. *(Classical Fenichel Theorem)*
Let $X_\varepsilon(x,y) = (f(x,y,\varepsilon), \varepsilon g(x,y,\varepsilon))$ be a family of vector fields given by a C^r-fast–slow vector field ($r \geq 2$). Let X_R be the slow flow. Let $S_0 \subset C_0$ be an n-dimensional properly embedded C^r-submanifold of $\mathbb{R}^{m+n}$. Suppose that S_0 is normally hyperbolic and has m^s eigenvalues with negative real part and m^u eigenvalues with positive real part. Then, for ε sufficiently small,

- *(i) There exists a family of C^r-manifolds S_ε such that S_ε is a normally hyperbolic locally invariant manifold for X_ε.*
- *(ii) There exist families of $(n+m^s)$- and $(n+m^u)$-dimensional local stable and unstable manifolds $W^s_{loc}(S_\varepsilon)$ and $W^u_{loc}(S_\varepsilon)$.*
- *(iii) All previously mentioned families of manifolds are C^{r-1} in ε.*

[1] C. KUEHN, *Multiple Time Scale Dynamics*, Chapter 2: *General Fenichel Theory* and Chapter 3: *Geometric Singular Perturbation Theory*, Springer, **2015**.

- *(iv) The vector field given by*

$$X_{C,\varepsilon}(p) = \begin{cases} \frac{1}{\varepsilon} X_\varepsilon(p) & \text{if } \varepsilon \neq 0 \text{ and } p \in S_\varepsilon \\ X_R(p) & \text{if } \varepsilon = 0 \text{ and } p \in S_0 \end{cases}$$

is a C^r-vector field on S_ε and $X_{C,\varepsilon}$ is a C^{r-1} family of vector fields in ε.

A.2 Fenichel Theorems Applied to Fast-Slow Systems

In this project, we were particularly interested in the application of this theory in the context of fast-slow systems that appear in certain topics of physics and biology, among others. In relation to fast-slow systems, the basic results of Fenichel Theory can be grouped into three main theorems. Emphasis has been placed on the first theorem and also the second theorem has been stated. There is also a third theorem that is worth mentioning in the context of fast-slow systems. Of course, geometric singular perturbation theory is extensive and more results could be cited. As a summary, we will state here the three aforementioned theorems.

A.2.1 First Fenichel Theorem

Theorem A.2. *(First Fenichel Theorem applied to fast-slow systems).*
Let S_0 be a compact normally hyperbolic submanifold of the critical manifold C_0. Let $f : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^m$ and $g : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$ be C^r-smooth functions, where $1 \leq r < \infty$. Then, for a sufficiently small $\varepsilon > 0$, the following hold:

- *(i) There exists a locally invariant manifold S_ε (that is, trajectories can enter or leave S_ε only through its boundary) diffeomorphic to S_0.*
- *(ii) As $\varepsilon \to 0$, S_ε has Hausdorff distance $O(\varepsilon)$ from S_0.*
- *(iii) As $\varepsilon \to 0$, the flow on S_ε converges to the slow flow.*
- *(iv) S_ε is C^r-smooth.*

A.2.2 Second Fenichel Theorem

Theorem A.3. *(Second Fenichel Theorem applied to fast-slow systems)*
Let S_0 be a compact normally hyperbolic submanifold of the critical manifold C_0. Let $f : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^m$ and $g : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$ be C^r-smooth functions, where $1 \leq r < \infty$. Then, instead of referring to S_0 and S_ε, all the statements of the First Fenichel Theorem also hold for their associated local manifolds. That is,

- *(i) S_0 can be replaced by $W^s_{loc}(S_0)$ or $W^u_{loc}(S_0)$.*
- *(ii) S_ε can be replaced by $W^s_{loc}(S_\varepsilon)$ or $W^u_{loc}(S_\varepsilon)$.*

A.2.3 Third Fenichel Theorem

Theorem A.4. *(Third Fenichel Theorem applied to fast-slow systems)*
Let S_0 be a compact normally hyperbolic submanifold of the critical manifold C_0. Let $f : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^m$ and $g : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$ be C^r-smooth functions, where $1 \leq r < \infty$. Then, there exists a manifold $F^u(p)$ for each $p \in S_0$ such that

- *(i) $\bigcup_{p \in S_0} F^u(p) = W^u_{loc}(S_0)$.*

- *(ii)* $F^u(p) \cap F^u(p') = \emptyset$ *for* $p \neq p'$.
- *(iii) For every* $t \geq 0$*, we have* $\phi_{-t}(F^u(p)) \subseteq F^u(\phi_{-t}(p))$.
- *(iv)* $F^u(p)$ *is tangent to* N_p^u *at* p*, being* N_p^u *the unstable component of the normal direction to* S_0*, i.e., a fast direction.*
- *(v) For every* $t \geq 0$*, there exist constants* $C_u > 0$ *and* $\lambda_u > 0$ *such that, if* $q \in F^u(p)$*, then*

$$||\phi_{-t}(p) - \phi_{-t}(q)|| < C_u e^{-\lambda_u t},$$

  where ϕ *is the flow.*
- *(vi)* $F^u(p)$ *is* C^r *with respect to the base point* p.

Equally, the same statements hold for manifolds $F^s(p)$ *replacing the u superscripts (for unstable) by s superscripts (for stable) and replacing* $-t$ *by* t *in statement (iii) (i.e.,* $F^u(p)$ *is negatively invariant and* $F^s(p)$ *is positively invariant).*

B Further Comments on the Karma Model

In the thesis, we have considered the following system:

$$\begin{cases} E' = -E + 2\left(E^* - n^M\right)\left(E^2 - \frac{1}{4}E^3\right), \\ n' = \varepsilon\left(\frac{1}{n_B}\theta(E-1) - n\right). \end{cases} \tag{5}$$

What is the effect of adding an external current in the equation for the evolution of the potential E?

B.1 Karma Model with External Current

For example, consider the following variation of the Karma Model with fast time derivatives:

$$\begin{cases} E' = -E + 2\left(E^* - n^M\right)\left(E^2 - \frac{1}{4}E^3\right) + I, \\ n' = \varepsilon\left(\frac{1}{n_B}\theta(E-1) - n\right), \end{cases} \tag{6}$$

where $I > 0$ is a dimensionless external current that is added to the system.

This creates important changes in the shape of the critical manifold.

Proposition B.1. *The critical manifold of the ODE Karma Model with external current is*

$$C_0 = \left\{ (E,n) \in \mathbb{R}^2 \;\middle|\; n = \sqrt[M]{E^* - \frac{E-I}{2E^2\left(1 - \frac{1}{4}E\right)}} \right\}. \tag{7}$$

Proof. Solve for n in $-E + 2\left(E^* - n^M\right)\left(E^2 - \frac{1}{4}E^3\right) + I = 0$. □

Now, the critical manifold does not have disconnected parts. As I increases, the behavior of the system and the presence of certain dynamical structures change. In that sense, I could be analyzed as a bifurcation parameter and find different ranges with different qualitative behavior. For example, consider the following two simulations with $I = 0.1$ and $I = 0.5$.

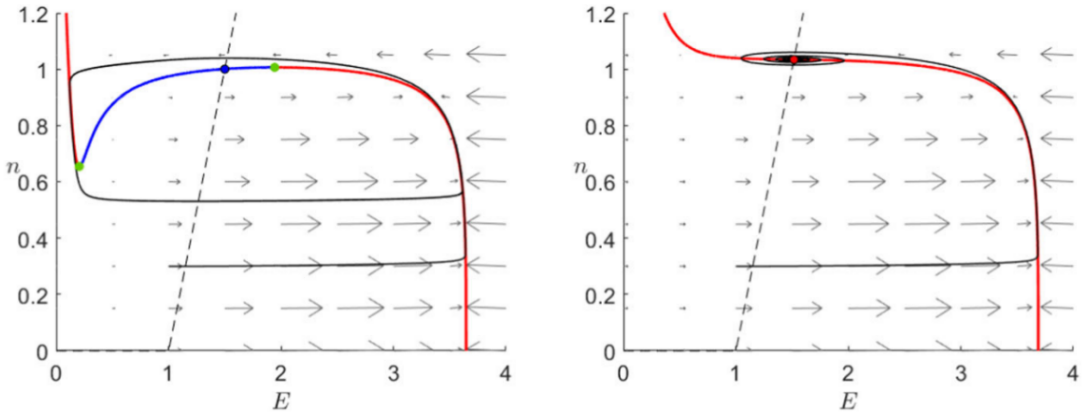


Figure 1: Karma Model simulations with external current $I = 0.1$ (left) and $I = 0.5$ (right). In red, attracting branches of the critical manifold. In blue, repelling branches of the critical manifold. In black, a typical trajectory (extracted from *Reduced Models of Cardiomyocytes Excitability: Comparing Karma and FitzHugh–Nagumo* by M. E. Gonzalez Herrero, C. Kuehn and K. Tsaneva-Atanasova cited in the thesis).

In the first case, we see that now there are two fold points separating the branches of the critical manifold (previously one) and only one equilibrium point (previously three). Relaxation oscillations appear.

In the second case, there are no fold points because the critical manifold is attracting everywhere and there is also an equilibrium point which may be of different type. Now, the trajectory is spiraling towards a stable equilibrium point.
Furthermore, an extension of the Karma Model which has been analyzed in the literature[2] consists of an extended system of the two dynamical variables from the standard system plus the consideration of I as an additional dynamical variable. This leads to a more complex model with three differential equations for E', n' and I', which is an interesting possibility of study if constant external currents fall short in certain attempts of modeling phenomena.

[2]M. E. GONZALEZ HERRERO, C. KUEHN AND K. TSANEVA-ATANASOVA, *Reduced Models of Cardiomyocytes Excitability: Comparing Karma and FitzHugh–Nagumo*, Bulletin of Mathematical Biology 83, 88, **2021**

C Further Comments on the Reduced Luo-Rudy Model

C.1 Lower Values of G_K

We have already worked with $G_K = 0.05$ mS/cm^2, $G_K = 0.045$ mS/cm^2 and $G_K = 0.04$ mS/cm^2. We observed different dynamical phenomena depending on such conductance. Most remarkably, the different number of extra spikes in the plateau associated to EADs and how this is related to the orbits and canards. However, for even lower values of G_K, new dynamical phenomena arise and it is worth dedicating a short extension on this matter.

Let us consider two new values: $G_K = 0.035$ mS/cm^2 and $G_K = 0.03345$ mS/cm^2. We will perform the corresponding numerical simulations yielding phase portraits and time series. As in the thesis, we will choose the initial condition $(V(0), x(0), f(0)) = (0, 0.5, 0.5)$. In order to visualize correctly the full phenomena, we will now simulate 12000 milliseconds for $G_K = 0.035$ mS/cm^2 and 16000 milliseconds for $G_K = 0.03345$ mS/cm^2.

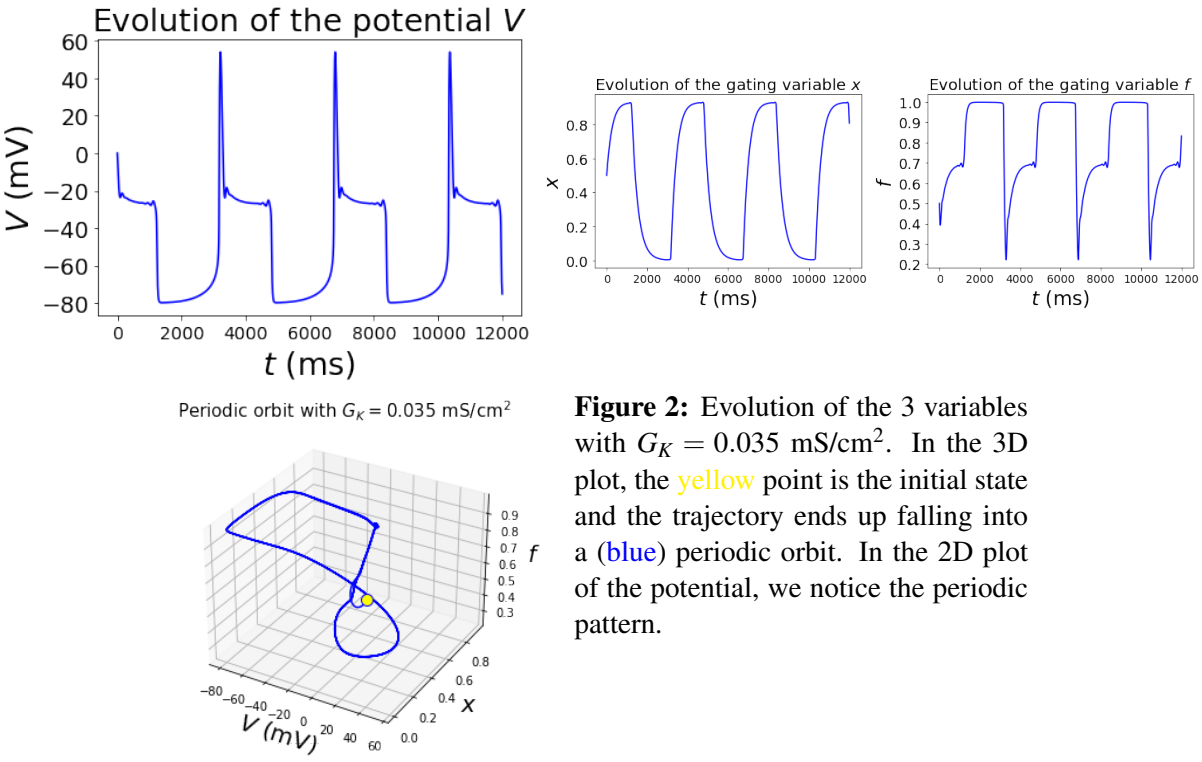


Figure 2: Evolution of the 3 variables with $G_K = 0.035$ mS/cm^2. In the 3D plot, the yellow point is the initial state and the trajectory ends up falling into a (blue) periodic orbit. In the 2D plot of the potential, we notice the periodic pattern.

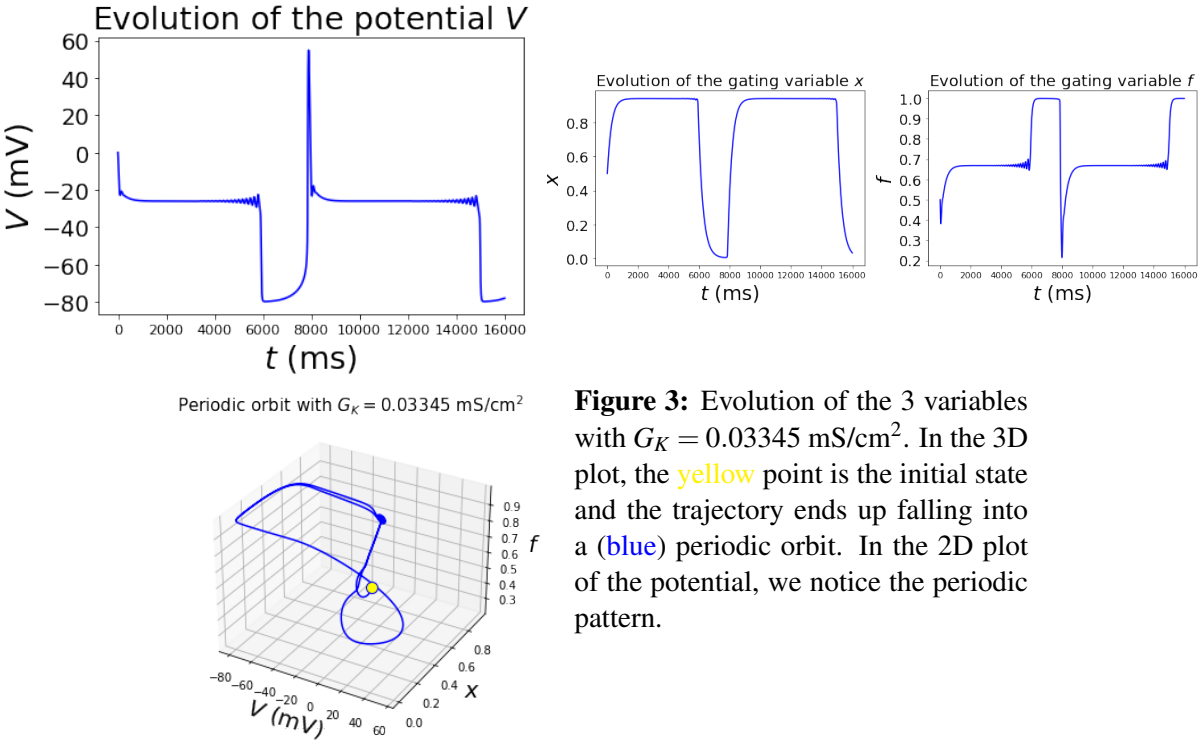


Figure 3: Evolution of the 3 variables with $G_K = 0.03345$ mS/cm^2. In the 3D plot, the yellow point is the initial state and the trajectory ends up falling into a (blue) periodic orbit. In the 2D plot of the potential, we notice the periodic pattern.

A clearly visible difference with respect to higher values of G_K is that the plateau zone is widened considerably for lower values of G_K. More oscillations appear in the plateau leading to a higher amount of EADs. We will also show the full 3D representation with the relevant dynamical objects:

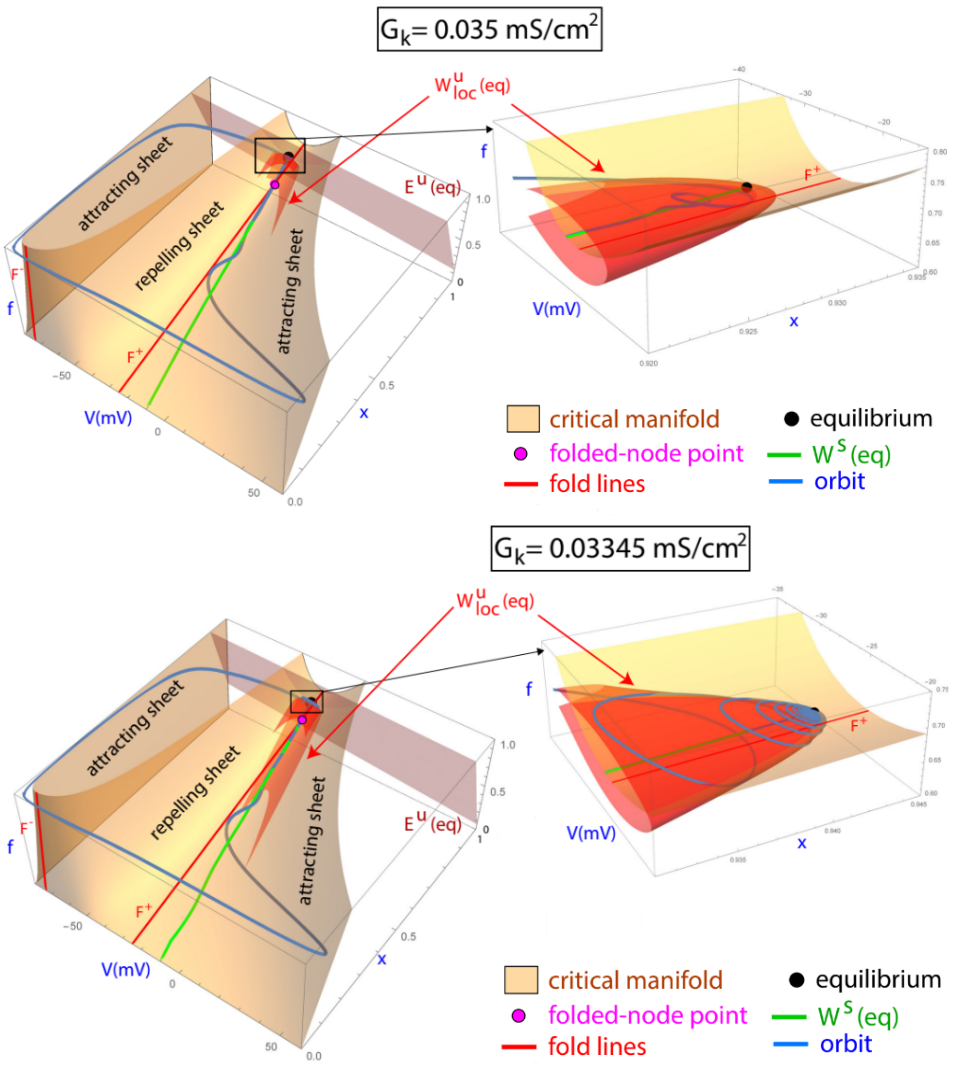


Figure 4: Relevant dynamical objects for $G_K = 0.035$ mS/cm^2 and $G_K = 0.03345$ mS/cm^2 (extracted from *Bifurcations and Slow-Fast Analysis in a Cardiac Cell Model for Investigation of Early Afterdepolarizations* by R. Barrio, M. A. Martínez, L. Pérez and E. Pueyo cited in the thesis).

In relation to the cases $G_K = 0.05,\ 0.045,\ 0.04$ mS/cm^2 that were examined in the thesis, we notice new dynamical phenomena. The equilibrium point now becomes a saddle focus, because now eigenvalues have non-zero imaginary part. The associated jacobian has one eigenvalue with negative real part and two eigenvalues with positive real part. The orbit oscillates (along with the canard phenomena) around the 1-dimensional stable manifold associated to the equilibrium point and it gets very close to the equilibrium point. Then, the orbit spirals outward, which can be seen with the 2-dimensional unstable manifold associated to the equilibrium point.

C.2 Addition of a Stimulus Current

An additional stimulus current I_{sti} could be added in the equation for the potential leading to the following system:

$$\begin{cases} V' = -\frac{I_{Ca}+I_K+I_{sti}}{C_m}, \\ f' = \frac{f_\infty(V)-f}{\tau_f}, \\ x' = \frac{x_\infty(V)-x}{\tau_x}. \end{cases} \tag{8}$$

As an example, let us consider the case with $G_K = 0.04$ mS/cm^2 and the usual initial condition $(V(0), x(0), f(0)) = (0, 0.5, 0.5)$. We will simulate 20000 milliseconds.
First consider $I_{sti} = 0.00331$ μA/cm^2.

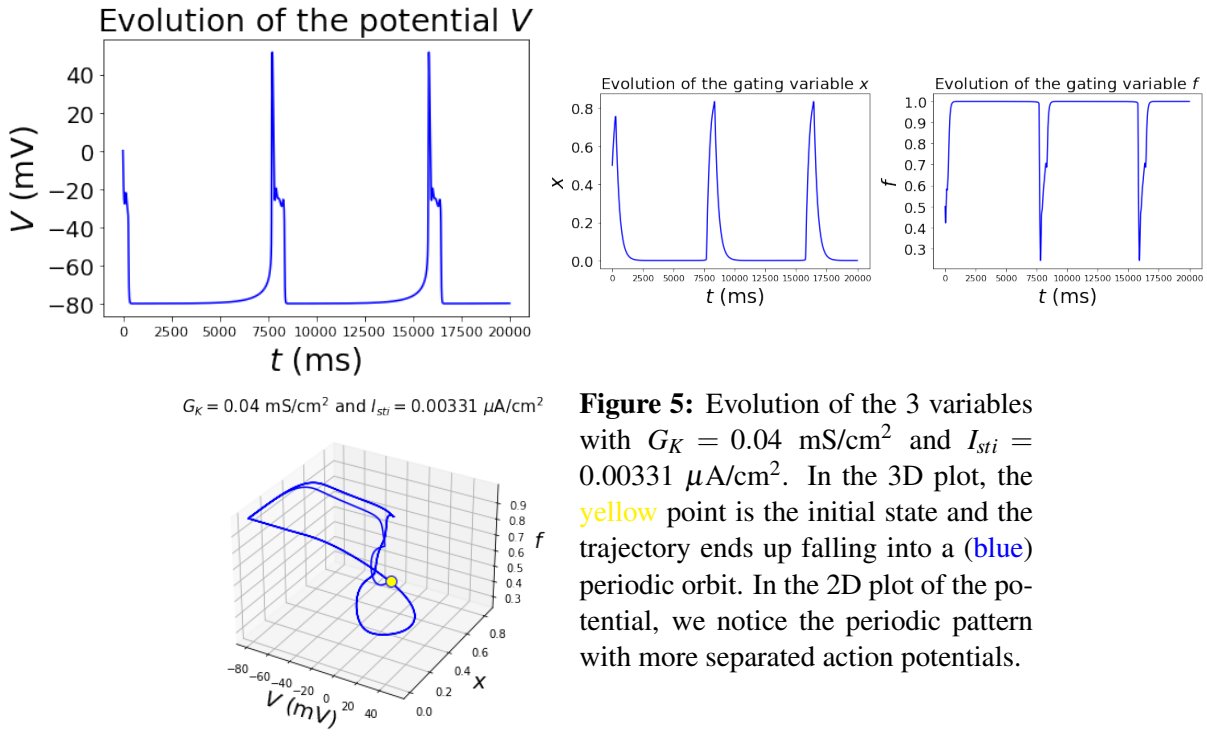


Figure 5: Evolution of the 3 variables with $G_K = 0.04$ mS/cm^2 and $I_{sti} = 0.00331$ μA/cm^2. In the 3D plot, the yellow point is the initial state and the trajectory ends up falling into a (blue) periodic orbit. In the 2D plot of the potential, we notice the periodic pattern with more separated action potentials.

For a sufficiently small stimulus current, the shape of the orbits is very similar to the usual dynamics that we have observed before. However, the orbits are traversed at a slower speed. We see that the action potentials have a larger separation. That is, the diastolic interval increases.

In contrast, let us observe what happens if we increase the current a bit.
Consider $I_{sti} = 0.00332$ μA/cm^2.

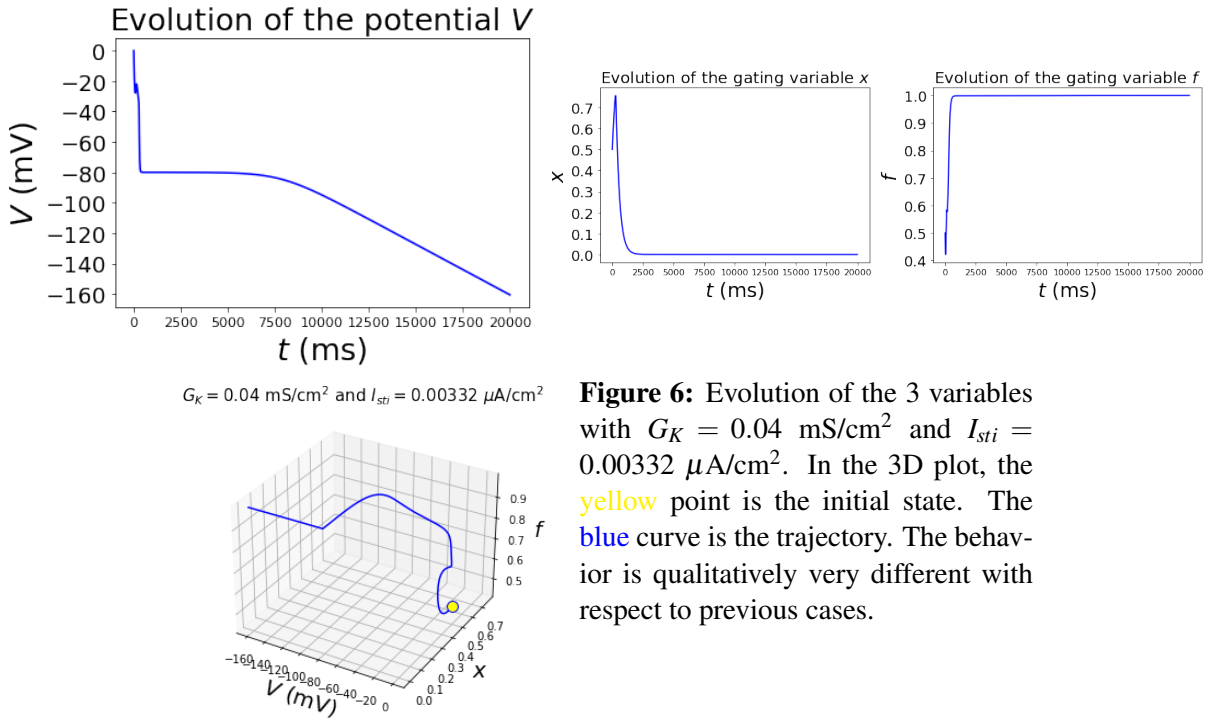


Figure 6: Evolution of the 3 variables with $G_K = 0.04$ mS/cm^2 and $I_{sti} = 0.00332$ μA/cm^2. In the 3D plot, the yellow point is the initial state. The blue curve is the trajectory. The behavior is qualitatively very different with respect to previous cases.

There is a change in the usual dynamics above a certain value of I_{sti}. The resulting trajectory is not a closed orbit. The trajectory converges to a line of constant $x = 0$ and $f = 1$ where the voltage V keeps decreasing to extremely negative values.

Let us now consider negative stimulus currents. Other types of interesting dynamics arise. For example, consider $I_{sti} = -0.275$ μA/cm^2 in a simulation of 30000 milliseconds for a better visualization.

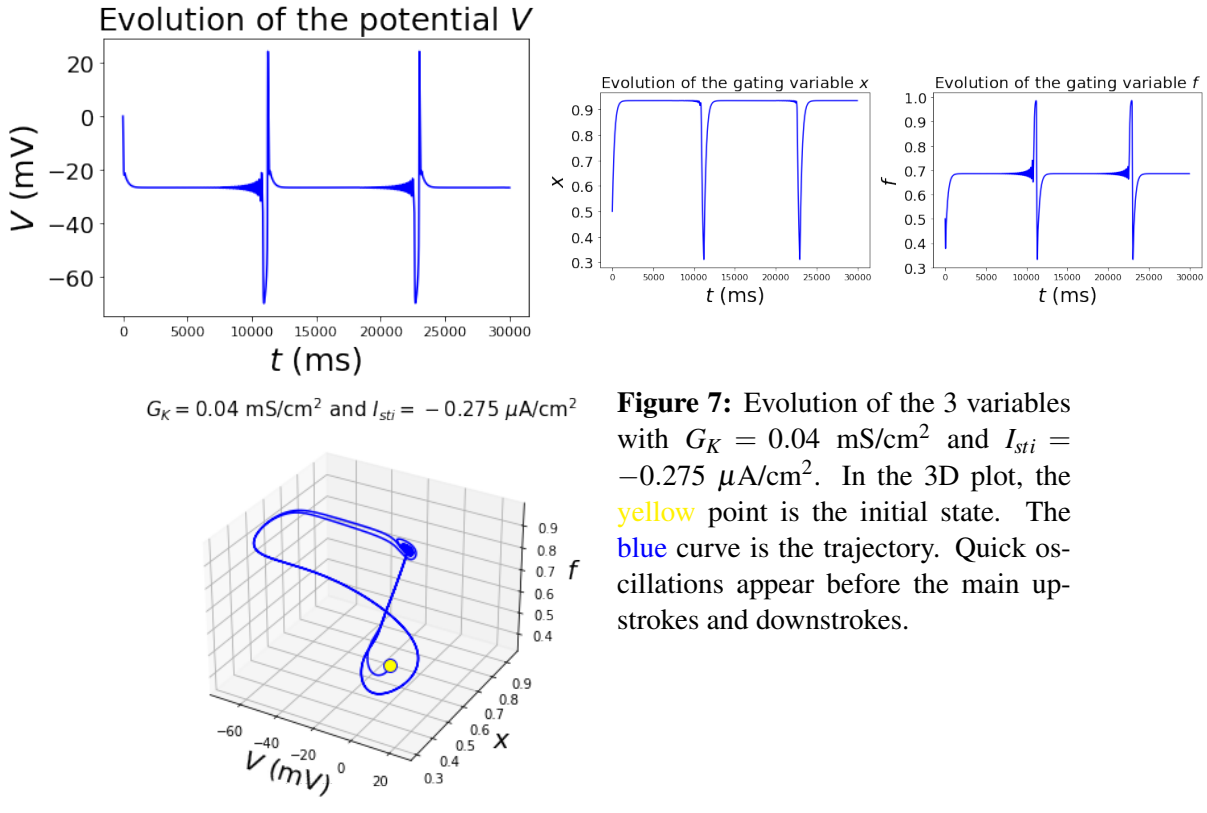


Figure 7: Evolution of the 3 variables with $G_K = 0.04$ mS/cm^2 and $I_{sti} = -0.275$ μA/cm^2. In the 3D plot, the yellow point is the initial state. The blue curve is the trajectory. Quick oscillations appear before the main upstrokes and downstrokes.

A more in-depth study could lead to meaningful interpretations of these phenomena and the characterization of the different regions of interest could be a topic for future work.

C.3 Bifurcation Analysis

First, we introduce basic definitions. The main reference is Kuznetsov's book[3].

Definition C.1. A **bifurcation** is the appearance of a topologically nonequivalent phase portrait under variation of parameters, i.e., a change of the topological type of the system as its parameters pass through a certain value.

There are several types of bifurcations, but here we will state some of the most typical ones and those who also appear in the model under consideration.

Definition C.2. Given a periodic orbit of the system, a *period-doubling bifurcation* is the type of bifurcation associated to the appearance of a new periodic orbit whose period is twice that of the original orbit.

Definition C.3. A *fold bifurcation* (also known as *saddle-node bifurcation*) is the type of local bifurcation associated to the collision and annihilation of two equilibrium points. Specifically, this is connected to the appearance of a zero eigenvalue in the jacobian of the system evaluated at the unique equilibrium point that is formed after the collision. This can occur in systems of dimension 1 or higher.

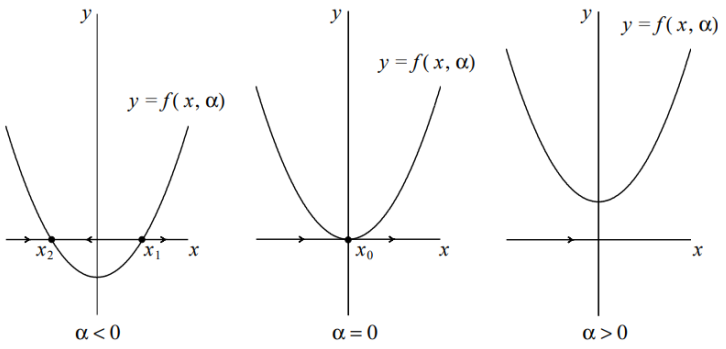


Figure 8: The 1-dimensional system $x' = x^2 + \alpha$ has a fold bifurcation at $\alpha = 0$ where the two equilibria $x = \pm\sqrt{-\alpha}$ collide (extracted from Kuznetsov's *Elements of Applied Bifurcation Theory*).

Definition C.4. A *Hopf bifurcation* (also known as *Poincaré–Andronov–Hopf bifurcation*) is the type of local bifurcation associated to the creation or disappearance of a periodic orbit (a limit cycle) as a result of a change in the stability of an equilibrium point. Specifically, this is connected to the situation where two complex conjugate eigenvalues cross the imaginary axis in the complex plane. This can occur in systems of dimension 2 or higher.
When varying the parameter from lower values to higher values, two types of Hopf bifurcations may arise:

- *Supercritical*: If a stable limit cycle appears at the critical parameter value and a limit cycle still exists for higher values.

[3] Y. A. KUZNETSOV, *Elements of Applied Bifurcation Theory*, Springer, **1998**.

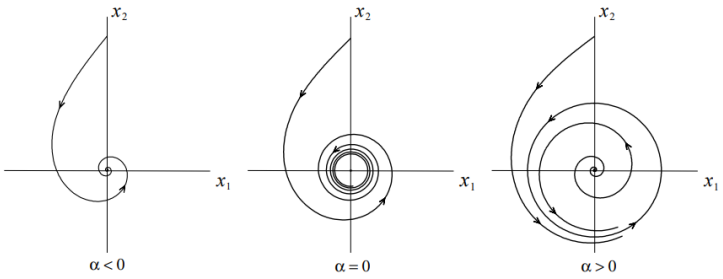


Figure 9: The 2-dimensional system $x_1' = \alpha x_1 - x_2 - x_1(x_1^2 + x_2^2)$, $x_2' = x_1 + \alpha x_2 - x_2(x_1^2 + x_2^2)$ has a supercritical Hopf bifurcation at $\alpha = 0$ (extracted from Kuznetsov's *Elements of Applied Bifurcation Theory*).

- *Subcritical*: If an unstable limit cycle exists for values of the parameter lower than the bifurcation value and the limit cycles disappear here for greater or equal values.

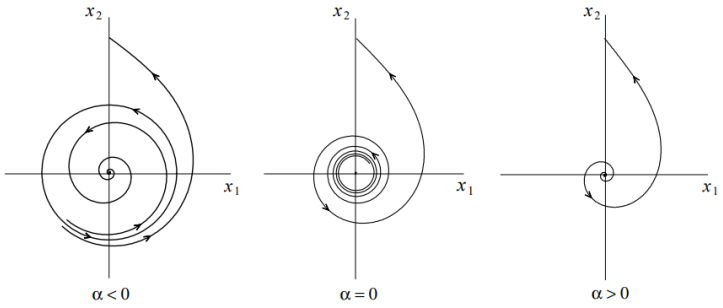


Figure 10: The 2-dimensional system $x_1' = \alpha x_1 - x_2 + x_1(x_1^2 + x_2^2)$, $x_2' = x_1 + \alpha x_2 + x_2(x_1^2 + x_2^2)$ has a subcritical Hopf bifurcation at $\alpha = 0$ (extracted from Kuznetsov's *Elements of Applied Bifurcation Theory*).

In the particular case of the Luo-Rudy Model[4], these bifurcations are identified. Since we have worked with variations of G_K, let us consider it as the bifurcation parameter. We can obtain the bifurcation diagram for the variation of the potential V with respect to the variation of G_K in the interval $[0.02, 0.46]$. First, let us consider the full interval to see the big picture.

[4]R. BARRIO, M. A. MARTÍNEZ, L. PÉREZ AND E. PUEYO, *Bifurcations and Slow-Fast Analysis in a Cardiac Cell Model for Investigation of Early Afterdepolarizations*, Mathematics, 8(6):880, **2020**

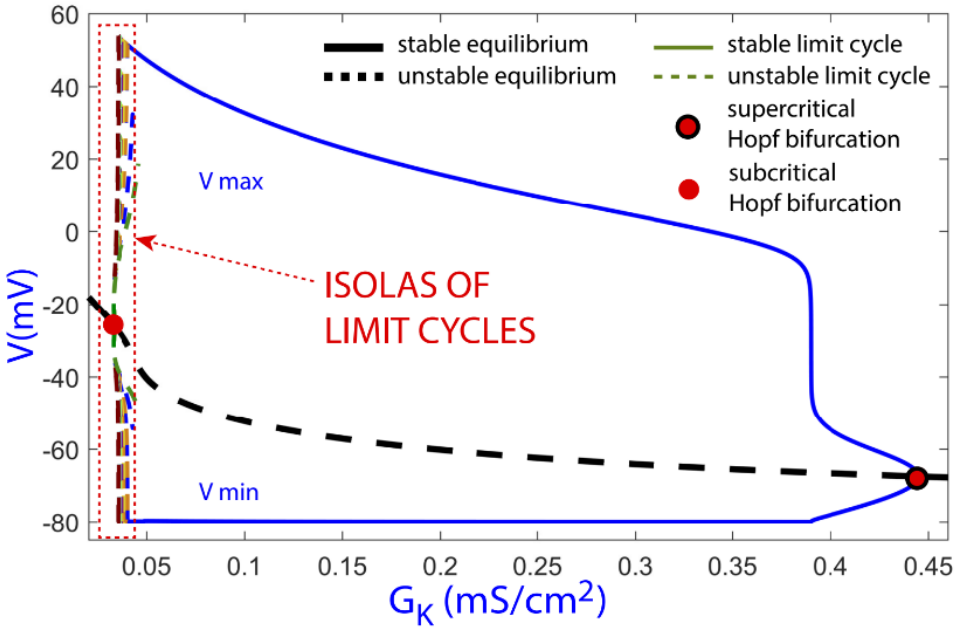


Figure 11: Bifurcation diagram with $G_K \in [0.02, 0.46]$. Supercritical and subcritical Hopf bifurcations are present. The color lines correspond to limit cycles (extracted from *Bifurcations and Slow-Fast Analysis in a Cardiac Cell Model for Investigation of Early Afterdepolarizations* by R. Barrio, M. A. Martínez, L. Pérez and E. Pueyo cited in the thesis).

In the small region on the left, there are EADs. We analyzed them for certain values of G_K. There also new dynamical objects called *isolas* of periodic orbits, i.e., simple closed curves of families of periodic orbits.
If we zoom in on this region, we can better visualize the position and nature of some of the bifurcations.

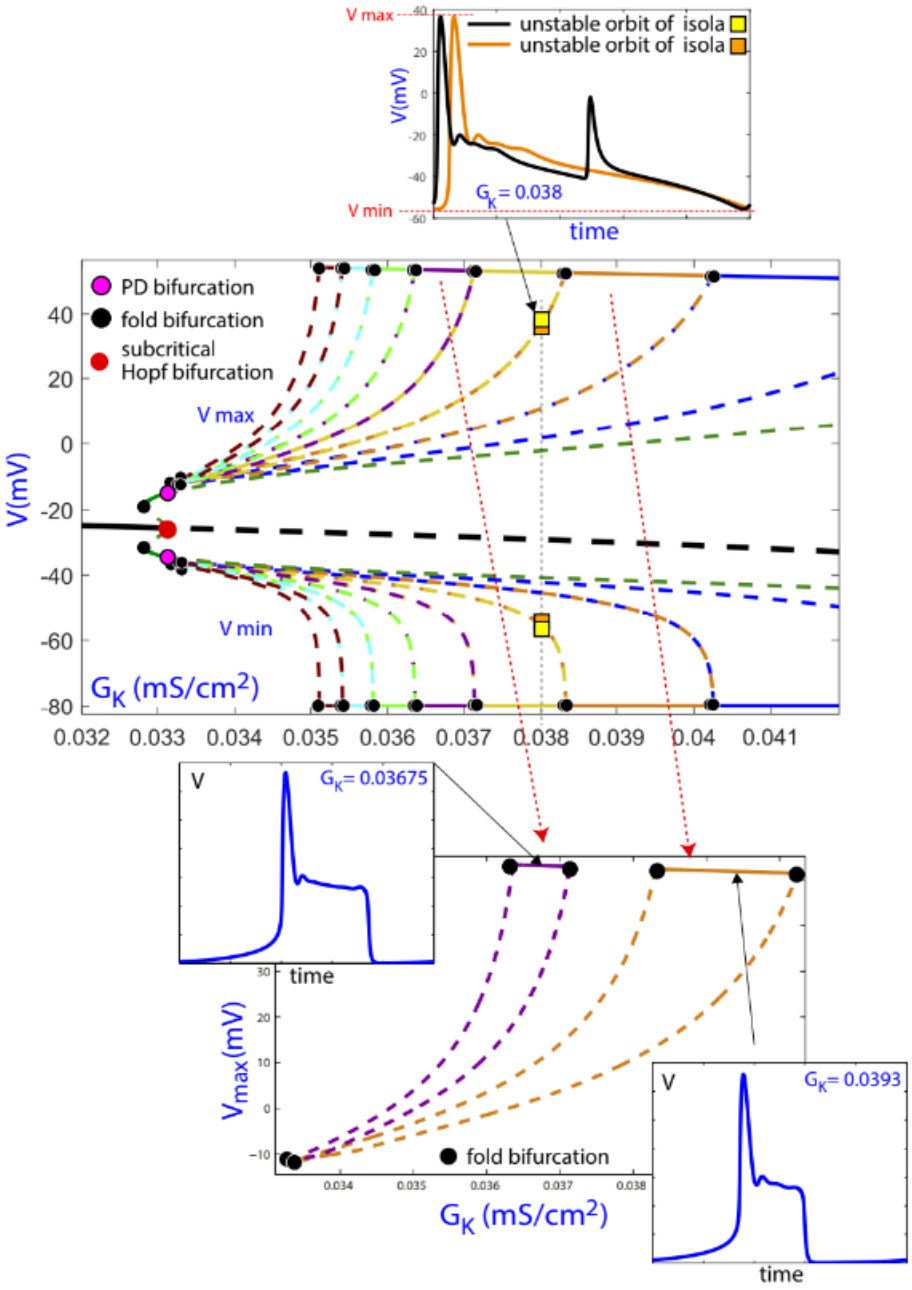


Figure 12: Magnification of the previous diagram. We see the position of period-doubling bifurcations, fold bifurcations and the subcritical Hopf bifurcation (extracted from *Bifurcations and Slow-Fast Analysis in a Cardiac Cell Model for Investigation of Early Afterdepolarizations* by R. Barrio, M. A. Martínez, L. Pérez and E. Pueyo cited in the thesis).

D Numerical Solution of Ordinary Differential Equations

In this project, I have carried out my simulations of systems of differential equations solving them numerically with the fourth order Runge-Kutta method.

D.1 Fourth Order Runge-Kutta Method

The idea was first conceived by Runge and presented in the paper *On the numerical solution of differential equations*[5] in 1895. Kutta published the paper *Contribution to the approximate integration of total differential equations*[6] in 1901. He classified fourth order methods and introduced the well-known method.
Given an initial value problem

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t,y), \qquad y(t_0) = y_0, \tag{9}$$

we want to solve it numerically in a certain interval by discretizing the time and obtaining a reasonable approximation of the unknown function at each discrete value of time.
We pick a step size $h > 0$ such that

$$t_n = t_{n-1} + h. \tag{10}$$

In the RK4 method, the values for the unknown function are given by

$$y_n = y_{n-1} + \frac{1}{6}h\left(k_{1,n-1} + 2k_{2,n-1} + 2k_{3,n-1} + k_{4,n-1}\right). \tag{11}$$

The variables k_1, k_2, k_3 and k_4 represent slopes in different parts of each interval:

- k_1 is a slope of y at the beginning of the interval.
- k_2 is a slope computed using y and k_1 at the midpoint of the interval.
- k_3 is a slope computed using y and k_2 at the midpoint of the interval.
- k_4 is a slope computed using y and k_3 at the end of the interval.

Their expressions are the following:

$$k_{1,n-1} = f\left(t_{n-1}, y_{n-1}\right), \tag{12}$$

$$k_{2,n-1} = f\left(t_{n-1} + \frac{1}{2}h, y_{n-1} + \frac{1}{2}hk_{1,n-1}\right), \tag{13}$$

$$k_{3,n-1} = f\left(t_{n-1} + \frac{1}{2}h, y_{n-1} + \frac{1}{2}hk_{2,n-1}\right), \tag{14}$$

$$k_{4,n-1} = f\left(t_{n-1} + h, y_{n-1} + hk_{3,n-1}\right). \tag{15}$$

[5] C. D. T. RUNGE, *Über die numerische Auflösing von Differentialgleichungen*, Mathematische Annalen, 46, p. 167-178, **1895**.

[6] M. W. KUTTA, *Beitrag zur näherungsweisen Integration totaler Differentialgleichungen*, Zeitschrift für Mathematik und Physik, 46, p. 435-453, **1901**.

E Code for Simulations and Plots

Some graphical results were extracted from the articles, books or web pages cited in the bibliography. Others were programmed by myself. I performed numerical simulations in *Python* for all the models implementing the fourth-order Runge-Kutta method. Also, useful functions from the `NumPy` and `Matplotlib` libraries were used for computations and plots. The parameters of the model can be modified accordingly to obtain the plots presented. Any of the following pieces of code can be run after loading the following libraries:

```
import numpy as np
import matplotlib.pyplot as plt
```

E.1 Van der Pol Oscillator Code

E.1.1 Simulation

```
d = 2 # Number of equations
ti = 0 # Initial time
tf = 5 # Final time
h = 0.0001 # Step size
epsilon = 0.04 # Relates fast and slow time scales in fast-slow systems

t = np.zeros((int((tf-ti)/h)+1 , 1)) # Time variable
y = np.zeros((int((tf-ti)/h)+1 , d)) # y[0] is "x" // y[1] is "y"

k1 = np.zeros((int((tf-ti)/h)+1 , d)) # Slope 1
k2 = np.zeros((int((tf-ti)/h)+1 , d)) # Slope 2
k3 = np.zeros((int((tf-ti)/h)+1 , d)) # Slope 3
k4 = np.zeros((int((tf-ti)/h)+1 , d)) # Slope 4

a = 0.5 #0.9948377923 #0.9934909321 # Control parameter

def equations(t,y):
    return [ (y[1]-y[0]**3/3+y[0])/epsilon , a-y[0] ]

#Initial conditions:

t0 = ti
y0 = [2,0]

t[0] = t0
y[0] = y0

# Runge-Kutta fourth-order method:

for i in range(1,int((tf-ti)/h)+1):

    k1[i-1] = equations(t[i-1] , y[i-1])
    k2[i-1] = equations(t[i-1] + h/2 , y[i-1] + h*k1[i-1]/2)
    k3[i-1] = equations(t[i-1] + h/2 , y[i-1] + h*k2[i-1]/2)
    k4[i-1] = equations(t[i-1] + h , y[i-1] + h*k3[i-1])

    y[i] = y[i-1] + h*(k1[i-1] + 2*k2[i-1] + 2*k3[i-1] + k4[i-1])/6
    t[i] = t[i-1] + h
```

E.1.2 Plots

```
def critical(x):
    return x**3/3 - x # Critical manifold

plt.figure(figsize=(8, 6))

x = np.linspace(-2.1, 2.1, 100)
plt.plot(x, critical(x), color='red')

xcoord = np.arange(-2, 2.2, 0.4)
ycoord = np.arange(-1, 1.5, 0.5)
X, Y = np.meshgrid(xcoord,ycoord)
xvelocity = (Y-X**3/3+X)/epsilon
plt.quiver(X, Y, xvelocity, np.zeros((xvelocity.shape[0]*xvelocity.shape[1])),
    color='gray', width=0.01)

plt.title('$a = $'+str(a), fontdict={'fontsize':25})
plt.xlabel('$x$', fontdict={'fontsize':25})
plt.ylabel('$y$', fontdict={'fontsize':25})
plt.xticks(fontsize=25)
plt.yticks(fontsize=25)

plt.plot([1], [-2/3], marker="o", markersize=10, markeredgecolor="red",
    markerfacecolor="green")
plt.plot([-1], [2/3], marker="o", markersize=10, markeredgecolor="red",
    markerfacecolor="green")

plt.plot(y[:,0],y[:,1], color='blue')
plt.show()
```

E.2 Karma Model Code

E.2.1 Simulation

```
d = 2 # Number of differential equations
ti = 0 # Initial time
tf = 600 # Final time
h = 0.01 # Time step

y = np.zeros((int((tf-ti)/h)+1 , d))
t = np.zeros((int((tf-ti)/h)+1 , 1))

k1 = np.zeros((int((tf-ti)/h)+1 , d)) # Slope 1
k2 = np.zeros((int((tf-ti)/h)+1 , d)) # Slope 2
k3 = np.zeros((int((tf-ti)/h)+1 , d)) # Slope 3
k4 = np.zeros((int((tf-ti)/h)+1 , d)) # Slope 4


def equations(t,y):

    # t is the independent fast time scale variable
    # y[0] is E, the transmembrane potential
    # y[1] is n, the slow gating variable for ion channels

    Estar = 1.5
    M = 4
    epsilon = 0.01
    nB = 0.5
    I = 0
```

```
26
27      #def stepfunction(E):
28      #     if E <= 0:
29      #         return 0
30      #     else:
31      #         return 1
32
33      def stepfunction(E):
34          return max(0,E)
35
36      equation1 = -y[0] + 2 * (Estar - y[1]**M) * (y[0]**2 - (y[0]**3)/4) + I
37      equation2 = epsilon * (stepfunction(y[0]-1)/nB - y[1])
38
39      return [equation1,equation2]
40
41  # Initial conditions:
42
43  t0 = ti
44  y0 = [0.5,0.5]
45
46  t[0] = t0
47  y[0] = y0
48
49  # Runge-Kutta fourth-order method:
50
51  for i in range(1,int((tf-ti)/h)+1):
52
53      k1[i-1] = equations(t[i-1] , y[i-1])
54      k2[i-1] = equations(t[i-1] + h/2 , y[i-1] + h*k1[i-1]/2)
55      k3[i-1] = equations(t[i-1] + h/2 , y[i-1] + h*k2[i-1]/2)
56      k4[i-1] = equations(t[i-1] + h , y[i-1] + h*k3[i-1])
57
58      y[i] = y[i-1] + h*(k1[i-1] + 2*k2[i-1] + 2*k3[i-1] + k4[i-1])/6
59      t[i] = t[i-1] + h
```

E.2.2 Plots

Evolution of the potential E:

```
1  plt.plot(t,y[:,0], color='blue')
2  plt.title('Evolution of the potential $E$', fontsize=25)
3  plt.xlabel('$t$', fontsize=25)
4  plt.ylabel('$E$', fontsize=25)
5  plt.xticks(fontsize=18)
6  plt.yticks(fontsize=18)
7  plt.show()
```

Evolution of the gating variable n:

```
1  plt.plot(t,y[:,1], color='blue')
2  plt.title('Evolution of the gating variable $n$', fontsize=20)
3  plt.xlabel('$t$', fontsize=25)
4  plt.ylabel('$n$', fontsize=25)
5  plt.xticks(fontsize=18)
6  plt.yticks(fontsize=18)
7  plt.show()
```

Phase portrait:

```
plt.plot(y[:,0],y[:,1], color='blue')

x = np.linspace(2-(2/3)*np.sqrt(6), 2+(2/3)*np.sqrt(6), 100)

def critical(x):
    return (1.5-(1/(2*x*(1-x/4))))**(1/4)

plt.plot(x, critical(x), color='red')
plt.vlines(x = 0, ymin = 0, ymax = 1, color = 'red')

plt.plot([2], [1], marker="o", markersize=10, markeredgecolor="red",
    markerfacecolor="lime")

plt.plot([0], [0], marker="o", markersize=10, markeredgecolor="red",
    markerfacecolor="black")
plt.plot([0.367], [0], marker="o", markersize=10, markeredgecolor="red",
    markerfacecolor="black")
plt.plot([1.496], [0.992], marker="o", markersize=10, markeredgecolor="red",
    markerfacecolor="black")

plt.title('Phase portrait of the Karma Model', fontdict={'fontsize':20})
plt.xlabel('$E$', fontdict={'fontsize':25})
plt.ylabel('$n$', fontdict={'fontsize':25})
plt.xticks(fontsize=15)
plt.yticks(fontsize=15)

x = np.linspace(0, 2.1, 100)

xcoord = np.arange(0, 2.1, 0.2)
ycoord = np.arange(0, 1.1, 0.2)
X, Y = np.meshgrid(xcoord,ycoord)
xvelocity = -X + 2 * (1.5 - Y**4) * (X**2 - (X**3)/4)
plt.quiver(X, Y, xvelocity, np.zeros((xvelocity.shape[0]*xvelocity.shape[1])),
    color='gray', width=0.01, scale=10)

plt.show()
```

E.3 Reduced Luo-Rudy Model Code

E.3.1 Simulation

```
d = 3 # Number of differential equations
ti = 0 # Initial time
tf = 8000 # Final time
h = 0.01 # Time step

y = np.zeros((int((tf-ti)/h)+1 , d))
t = np.zeros((int((tf-ti)/h)+1 , 1))

k1 = np.zeros((int((tf-ti)/h)+1 , d)) # Slope 1
k2 = np.zeros((int((tf-ti)/h)+1 , d)) # Slope 2
k3 = np.zeros((int((tf-ti)/h)+1 , d)) # Slope 3
k4 = np.zeros((int((tf-ti)/h)+1 , d)) # Slope 4


def equations(t,y):

    # t is the independent fast time scale variable
    # y[0] is V, the transmembrane potential
    # y[1] is f, the inactivation gating variable associated to the Ca current
```

```
    # y[2] is x, the activation gating variable associated to the K current

    Cm = 0.5
    tauf = 80
    taux = 300
    GCa = 0.025
    ECa = 100
    GK = 0.05
    EK = -80
    Ef = -20
    thetaf = 8.6
    Ex = -40
    thetax = 5
    Ed = -35
    thetad = 6.24
    Isti = 0

    def finfinity(V):
        return 1 / (1 + np.exp((V-Ef)/(thetaf)))

    def xinfinity(V):
        return 1 / (1 + np.exp((Ex-V)/(thetax)))

    def dinfinity(V):
        return 1 / (1 + np.exp((Ed-V)/(thetad)))

    def ICa(V,f):
        return GCa * dinfinity(V) * f * (V-ECa)

    def IK(V,x):
        return GK * x * (V-EK)

    equation1 = -(ICa(y[0],y[1])+IK(y[0],y[2])+Isti)/(Cm)
    equation2 = (finfinity(y[0])-y[1])/(tauf)
    equation3 = (xinfinity(y[0])-y[2])/(taux)

    return [equation1,equation2,equation3]

# Initial conditions:

t0 = ti
y0 = [0,0.5,0.5]

t[0] = t0
y[0] = y0

# Runge-Kutta fourth-order method:

for i in range(1,int((tf-ti)/h)+1):

    k1[i-1] = equations(t[i-1] , y[i-1])
    k2[i-1] = equations(t[i-1] + h/2 , y[i-1] + h*k1[i-1]/2)
    k3[i-1] = equations(t[i-1] + h/2 , y[i-1] + h*k2[i-1]/2)
    k4[i-1] = equations(t[i-1] + h , y[i-1] + h*k3[i-1])

    y[i] = y[i-1] + h*(k1[i-1] + 2*k2[i-1] + 2*k3[i-1] + k4[i-1])/6
    t[i] = t[i-1] + h
```

E.3.2 Plots

Evolution of the potential V:

```
plt.plot(t,y[:,0], color='blue')
plt.title('Evolution of the potential $V$', fontsize=25)
plt.xlabel('$t$ (ms)', fontsize=25)
plt.ylabel('$V$ (mV)', fontsize=25)
plt.xticks(fontsize=11)
plt.yticks(fontsize=18)
plt.show()
```

Evolution of the gating variable f:

```
plt.plot(t,y[:,1], color='blue')
plt.title('Evolution of the gating variable $f$', fontsize=20)
plt.xlabel('$t$ (ms)', fontsize=25)
plt.ylabel('$f$', fontsize=25)
plt.xticks(fontsize=11)
plt.yticks(fontsize=18)
plt.show()
```

Evolution of the gating variable x:

```
plt.plot(t,y[:,2], color='blue')
plt.title('Evolution of the gating variable $x$', fontsize=20)
plt.xlabel('$t$ (ms)', fontsize=25)
plt.ylabel('$x$', fontsize=25)
plt.xticks(fontsize=11)
plt.yticks(fontsize=18)
plt.show()
```

Phase portrait:

```
ax = plt.axes(projection='3d')
ax.plot(y[:,0],y[:,2],y[:,1], color='blue')
ax.figure.set_size_inches(5,5)
ax.set_title('Periodic orbit with $G_K = 0.05$ mS/cm$^2$', fontsize=15)
ax.set_xlabel('$V$ (mV)', fontsize=20)
ax.set_ylabel('$x$', fontsize=20)
ax.set_zlabel('$f$', fontsize=20)
plt.plot([0], [0.5], [0.5], marker="o", markersize=10, markeredgecolor="blue",
    markerfacecolor="yellow")
plt.show()
```